Week 98: Growing from a few seeds

Adult guide built from the problem files · Unpiloted; physically unrehearsed.

The answer keys, proofs, preparation and launch are in the print guide, week-98-facilitator.pdf, whose source is guide/facilitator.tex and guide/solutions.tex. This view lists each problem's numbers, route and idea.

Problems and routes

Problem	2–3	4–5	route
Three seeds that fill, three that get stuck	P1		first
How many final totals can three seeds make?		P1	first
Fewest seeds for a 4-by-4 board	P2	P2	first
Three seeds with finishes of 3, 4, 6 and 9	P3		return
The same seeds grown in different orders	P4		first
Fewest seeds for 5-by-5 and 10-by-10		P3	first
Exposed sides before and after an addition		P4	return
Can three seeds fill 4-by-4? Can four fill 5-by-5?		P5	return
Every seed needed, yet not the fewest	P5	P6	return

First route. Grades 2–3 P1–P2, P4; Grades 4–5 P1–P3. **Return visits.** Grades 2–3 P3, P5; Grades 4–5 P4–P6.

Key, one entry per problem

Three seeds that fill, three that get stuck

Grades 2–3 P1 · first route

Idea. The same number of seeds can fill the board or stop at once, depending on where they are.

How many final totals can three seeds make?

Grades 4–5 P1 · first route

Idea. Three seeds on 3 by 3 finish with exactly 3, 4, 6 or 9 filled squares.

Fewest seeds for a 4-by-4 board

Grades 2–3 P2 · Grades 4–5 P2 · first route

Idea. Four seeds fill 4 by 4 and three never do.

Three seeds with finishes of 3, 4, 6 and 9

Grades 2–3 P3 · return route

Idea. Where three seeds sit decides the size of the finish.

The same seeds grown in different orders

Grades 2–3 P4 · first route

Idea. Every complete run from the same seeds ends with the same filled squares.

Fewest seeds for 5-by-5 and 10-by-10

Grades 4–5 P3 · first route

Idea. The diagonal fills an n-by-n board with n seeds.

Exposed sides before and after an addition

Grades 4–5 P4 · return route

Idea. An addition touching k filled squares changes the exposed sides by 4 - 2k, never more than 0.

Can three seeds fill 4-by-4? Can four fill 5-by-5?

Grades 4–5 P5 · return route

Idea. Exposed sides never increase, so s seeds fill an n-by-n board only if s is at least n.

Every seed needed, yet not the fewest

Grades 2–3 P5 · Grades 4–5 P6 · return route · wording differs by band

Idea. A start can need every seed without being smallest: four indispensable seeds on 3 by 3, where three diagonal seeds fill.